Precursor-emission sensitivity in Stevenson et al. (2022)

Reference. Stevenson et al., *Atmospheric Chemistry and Physics* 22, 14243–14252. doi:10.5194/acp-22-14243-2022.

Method. The study applies sensitivities from the first Task Force on Hemispheric Transport of Air Pollution experiment to lockdown-related changes in NO_x, CO, and non-methane volatile organic compound (NMVOC) emissions. The underlying model experiments perturb 2001 anthropogenic emissions by 20%. Eleven models supply regional responses; four supply the global experiments needed to determine the rest-of-world contribution. Chemical lifetimes are converted to perturbation lifetimes using each model's methane feedback factor. The response after one year is obtained from the equilibrium methane response by multiplying by $1 - \exp(-1/\tau_p)$, in which τ_p is the perturbation lifetime in years. This represents a one-year forcing, not an instantaneous change in the methane loss flux. Extra aviation NO_x is added after accounting for the aviation contribution already present in the scaled experiments.

Findings. The four-model mean attributes 2.86 ± 1.13 ppb yr^{-1} of the 2020 methane growth-rate increase to precursor-emission reductions, or $54 \pm 21\%$ of the observed 5.3 ppb yr^{-1} anomaly. NO_x reductions contribute 4.75 ± 1.02 ppb yr^{-1}, offset by CO (-1.42 ± 0.29) and NMVOC (-0.48 ± 0.39) reductions. The additional aviation correction is 0.345 ppb yr^{-1}; the total aviation contribution, including what was already counted, is about 0.93 ppb yr^{-1}. Using all available regional models raises the combined central estimate to 3.16 ppb yr^{-1}.

Supplement and uncertainty. The supplementary workbook provides lifetimes, methane feedback factors, equilibrium responses, emissions, and one-year sensitivities. Table S13 reproduces the main attribution table, and its uncertainty combines species-wise model standard deviations in quadrature. These errors do not include uncertainty in the lockdown emission estimates or in transferring sensitivities from 2001 to 2020. Table S11 gives global surface NO_x, CO, and NMVOC changes of -19.38 Tg NO_2, -73.38 Tg CO, and -15.65 Tg C, respectively; aviation NO_x decreases by 0.83 Tg NO_2.

Use in this reconstruction. We extract the workbook's cached numerical values and retain its species and regional separation. We convert ppb to Tg using He et al.'s 2.77 Tg ppb^{-1} for a common comparison, explicitly retaining the distinction between a one-year burden response and $-m_{2019}\Delta k_{\mathrm{OH}}$. The study supplies neither an inferred methane-emission anomaly nor a complete 2019–2022 budget. We therefore compare its chemistry contribution separately and leave those missing quantities unfilled. Its perturbation lifetime is not substituted for He's chemical lifetime.